Clustering

Clustering

- A way of grouping the data points into different clusters, consisting of similar data points. The objects with the possible similarities remain in a group that has less or no similarities with another group.
- Cluster: A collection of data objects
  - similar (or related) to one another within the same group
  - dissimilar (or unrelated) to the objects in other groups
- Cluster analysis (or *clustering*, *data segmentation, …*)
  - Finding similarities between data according to the characteristics found in the data and grouping similar data objects into clusters

How the clustering Algorithm will works:

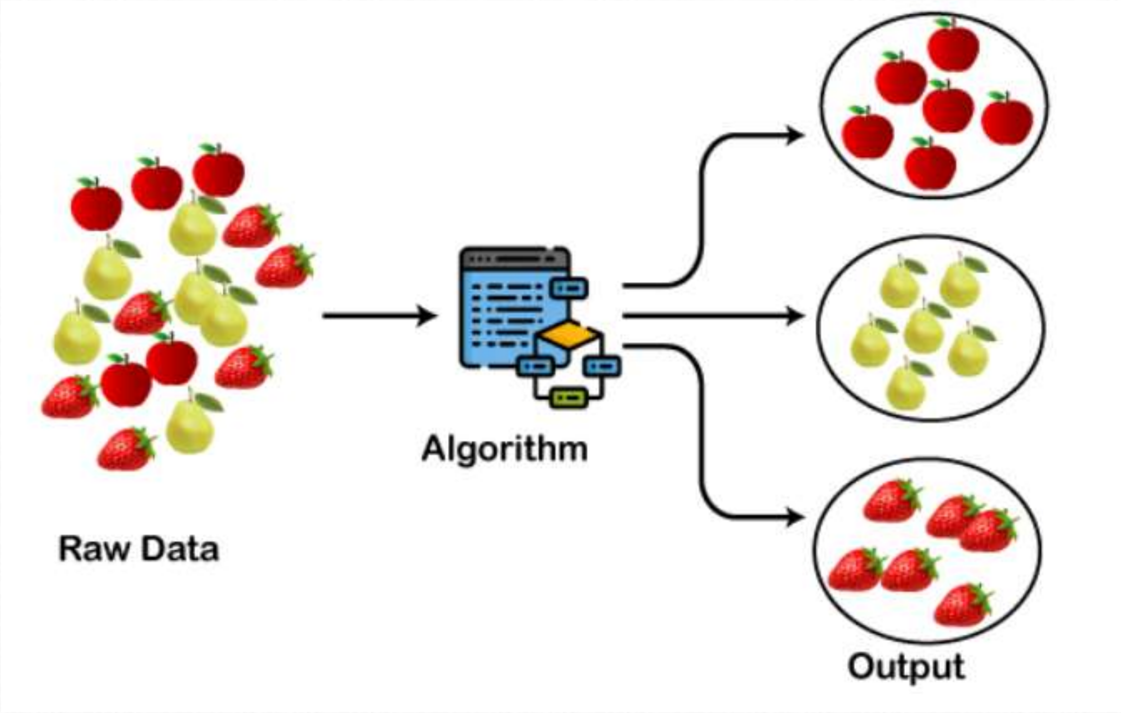


fig:The data from our customer base is divided into clusters, we can make an informed decision about who we think is best suited for this product .

Clustering for data understanding and Applications

- Biology: taxonomy of living things: kingdom, phylum, class, order, family, genus and species
- Information retrieval: document clustering
- Land use: Identification of areas of similar land use in an earth observation database
- Marketing: Help marketers discover distinct groups in their customer bases, and then use this knowledge to develop targeted marketing programs
- City-planning: Identifying groups of houses according to their house type, value, and geographical location
- Earth-quake studies: Observed earth quake epicenters should be clustered along continent faults
- Climate: understanding earth climate, find patterns of atmospheric and ocean
- Economic Science: market research.

Applications

- Market Segmentation
- Statistical data analysis
- Social network analysis
- Image segmentation
- Anomaly detection, etc

Quality: What Is Good Clustering

- A good clustering method will produce high quality clusters
- high intra-class similarity: cohesive within clusters
- low inter-class similarity: distinctive between clusters
- The quality of a clustering method depends on
  - the similarity measure used by the method
  - its implementation, and
  - Its ability to discover some or all of the hidden patterns

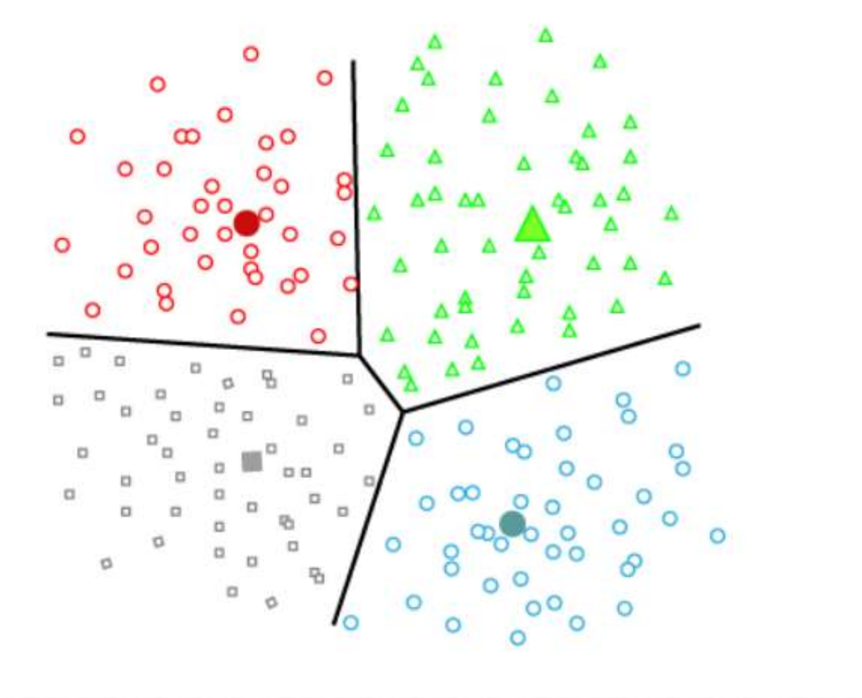
Major Clustering Approaches

- Partitioning approach:
  - Construct various partitions and then evaluate them by some criterion, e.g., minimizing the sum of square errors
  - Typical methods: k-means, k-medoids, CLARANS
- Hierarchical approach:
  - Create a hierarchical decomposition of the set of data (or objects) using some criterion
  - Typical methods: Diana, Agnes, BIRCH, CAMELEON
- Density-based approach:
  - Based on connectivity and density functions
  - Typical methods: DBSACN, OPTICS, DenClue

- Grid-based approach:
  - based on a multiple-level granularity structure
  - Typical methods: STING, WaveCluster, CLIQUE
- Model-based:
  - A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
  - Typical methods: EM, SOM, COBWEB
- Frequent pattern-based:
  - Based on the analysis of frequent patterns
  - Typical methods: p-Cluster

- User-guided or constraint-based:
  - Clustering by considering user-specified or application-specific constraints
  - Typical methods: COD (obstacles), constrained clustering
- Link-based clustering:
  - Objects are often linked together in various ways
  - Massive links can be used to cluster objects: SimRank, LinkClus

Partitioning Clustering



Introduction

- It divides the data into non-hierarchical groups.
- It is also known as the centroid-based method.
- The most common example of partitioning clustering is the K-Means Clustering algorithm.
- In this type, the dataset is divided into a set of k groups, where K is used to define the number of pre-defined groups.
- The cluster center is created in such a way that the distance between the data points of one cluster is minimum as compared to another cluster centroid.

- Partitioning a database D of n objects into a set of k clusters, such that the sum of squared distances is minimized (where ci is the centroid or medoid of cluster Ci.

$$E = \Sigma_{i=1}^{k} \Sigma_{p \in C_i} (p - c_i)^2$$

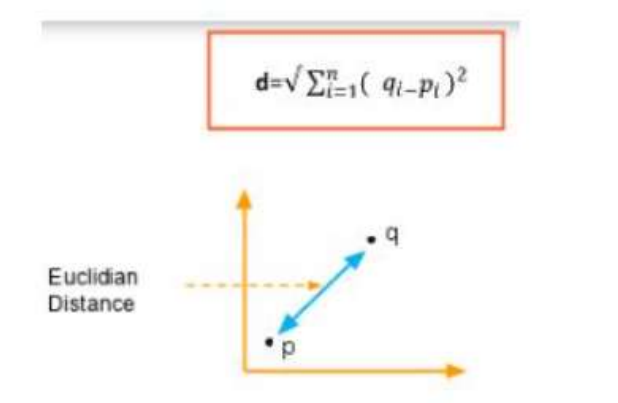
- Given k, find a partition of k clusters that optimizes the chosen partitioning criterion
- Global optimal: exhaustively enumerate all partitions
- Heuristic methods: k-means and k-medoids algorithms
- k-means (MacQueen'67, Lloyd'57/'82): Each cluster is represented by the center of the cluster
- k-medoids or PAM (Partition around medoids) (Kaufman & Rousseeuw'87): Each cluster is represented by one of the objects in the cluster

Distance between datapoints

- ❑ We assume that each data point is a n-dimensional vector
- ❑ K-Means clustering supports various kinds of distance measures, such as:
  1. Euclidean distance measure
  2. Manhattan distance measure

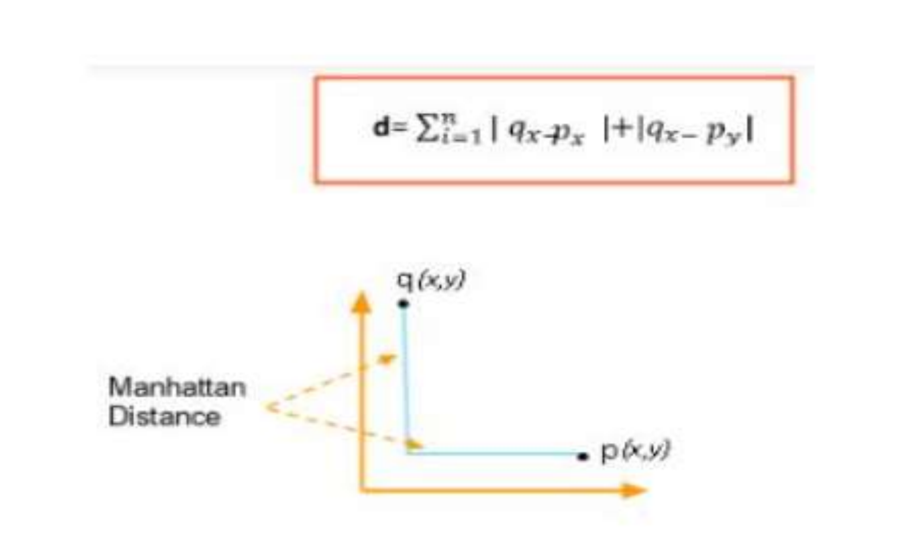
- Euclidean Distance Measure

❑ **If we have a point P and point Q, the Euclidean distance is an ordinary straight line.**



Manhattan Distance Measure

- The Manhattan distance is the simple sum of the horizontal and vertical components or the distance between two points measured along axes at right angles.

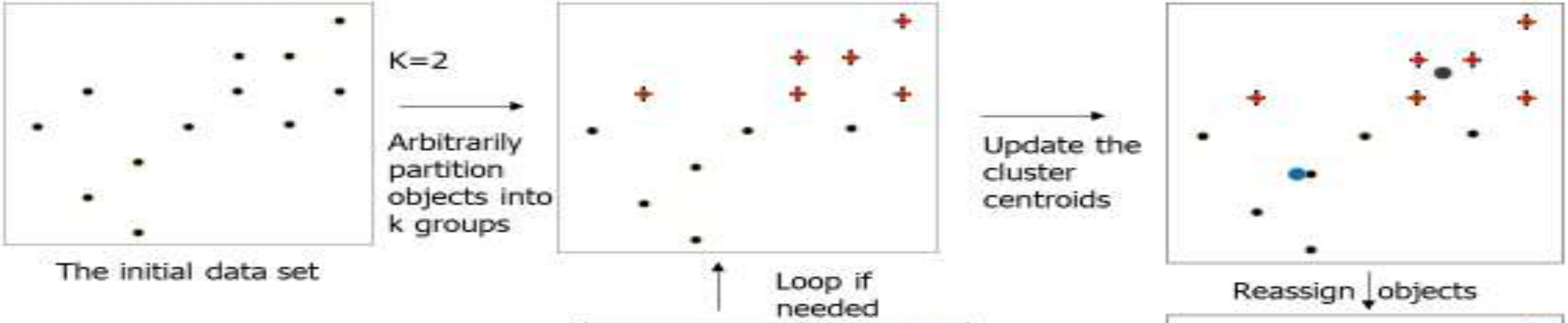


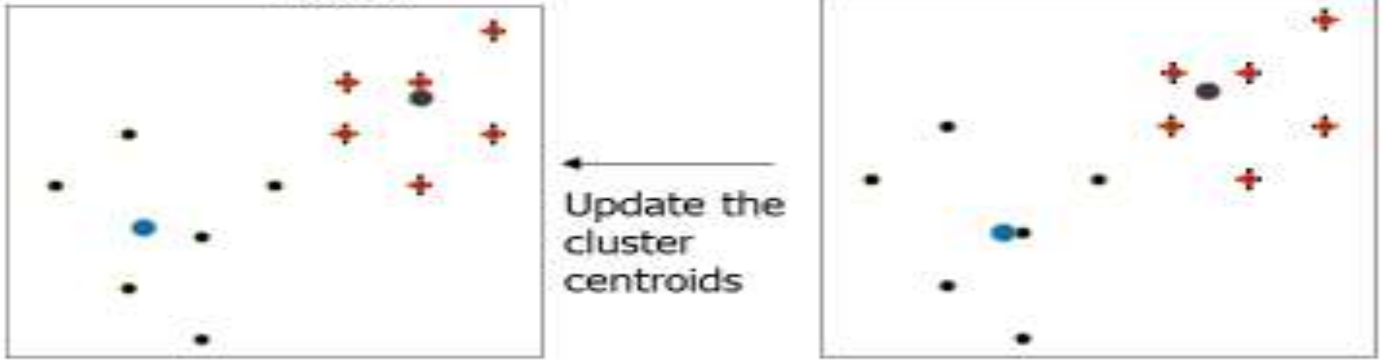
The K-Means Clustering Method

Given k, the k-means algorithm is implemented in four steps:

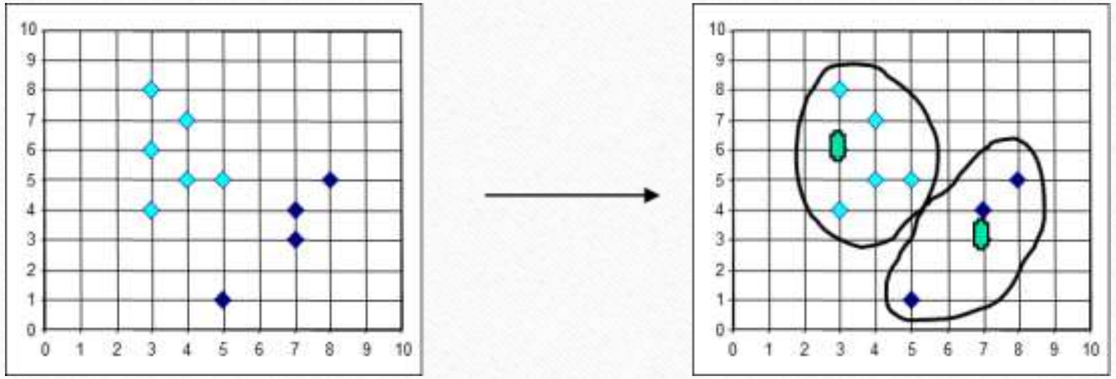
- Partition objects into k nonempty subsets
- Compute seed points as the centroids of the clusters of the current partitioning (the centroid is the center, i.e., mean point, of the cluster)
- Assign each object to the cluster with the nearest seed point
- Go back to Step 2, stop when the assignment does not change

An Example of *K-Means* Clustering





- Partition objects into *k* nonempty subsets
- Repeat
  - Compute centroid (i.e., mean point) for each partition
  - Assign each object to the cluster of its nearest centroid
- Until no change

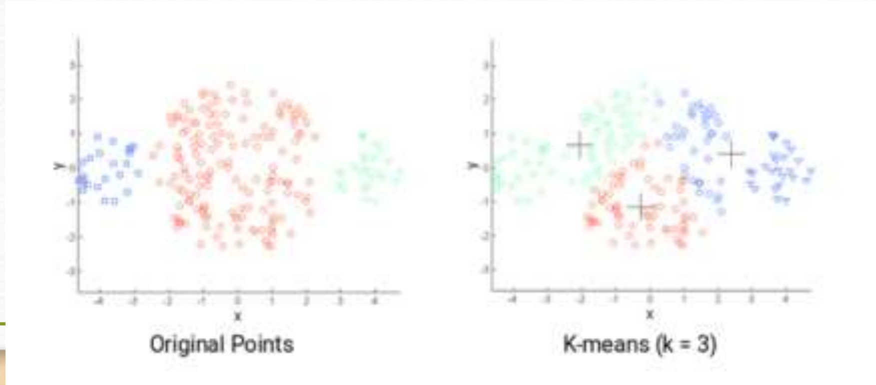


10 9 8 7 6 5 4 3 2 1 0
0 1 2 3 4 5 6 7 8 9 10
10 9 8 7 6 5 4 3 2 1 0
0 1 2 3 4 5 6 7 8 9 10

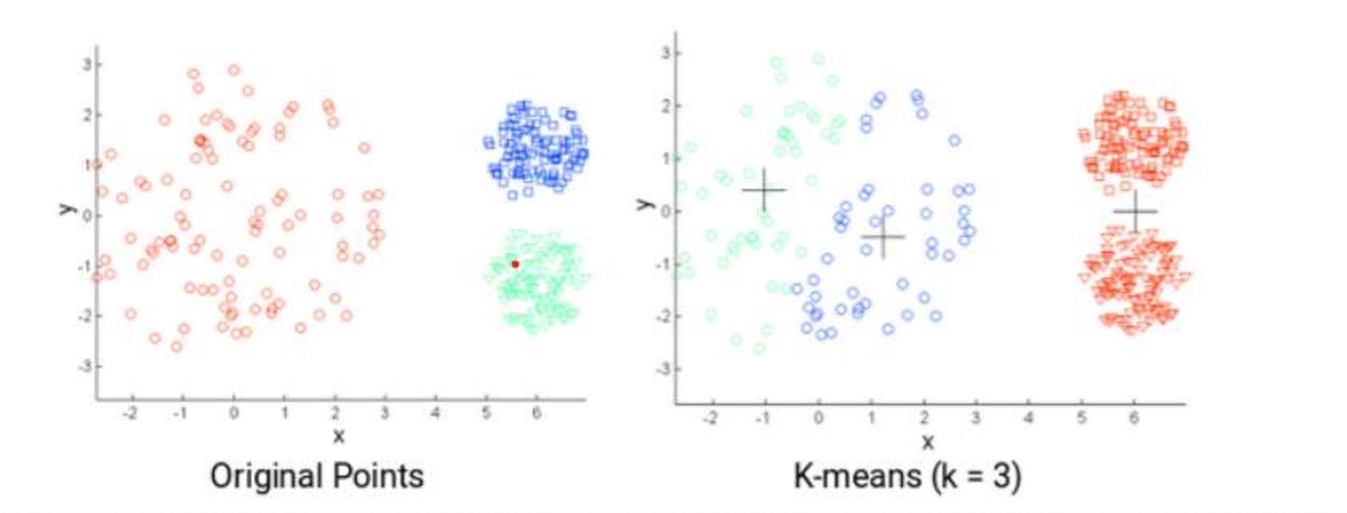
Challenges with the k-means clustering

- **Finding the optimal k value,** especially for noisy data. The appropriate value of k depends on the data structure and the problem being solved. It is important to choose the right value of k, as a small value can result in under-clustered data, and a large value can cause over-clustering.

- **The size of clusters is different.**



- The densities of the original points are different.



Question

- Use K-means clustering algorithm to decide the following data into two clusters.

x1	1	2	2	3	4	5
x2	1	1	3	2	3	5

- Step 1: Choose randomly two clusters:
  - V1=(2,1) and v2=(2,3)

Datapoint	Distance from v1	Distance from v2	Assigned center
a1(1,1)	1	2.24	V1
a2(2,1)	0	2	V1
a3(2,3)	2	0	V2
a4(3,2)	1.4	1.4	V1
a5(4,3)	2.83	2	V2
a6(5,5)	5	3.61	v2

- Step 3: Assign each datapoint to clusters
- cluster of v1 ={a1,a2,a4}
- cluster of v2 = {a3,a5,a6}
- Step 4: Recalculate the cluster center's
- v1=[a1+a2+a4]/3
- = (2,1.33)
- v2=[a3+a5+a6]/3
- = (3.67,3.67)

- Step 5: Repeat from step 2 until we get same cluster center or same cluster elements as in the previous iteration

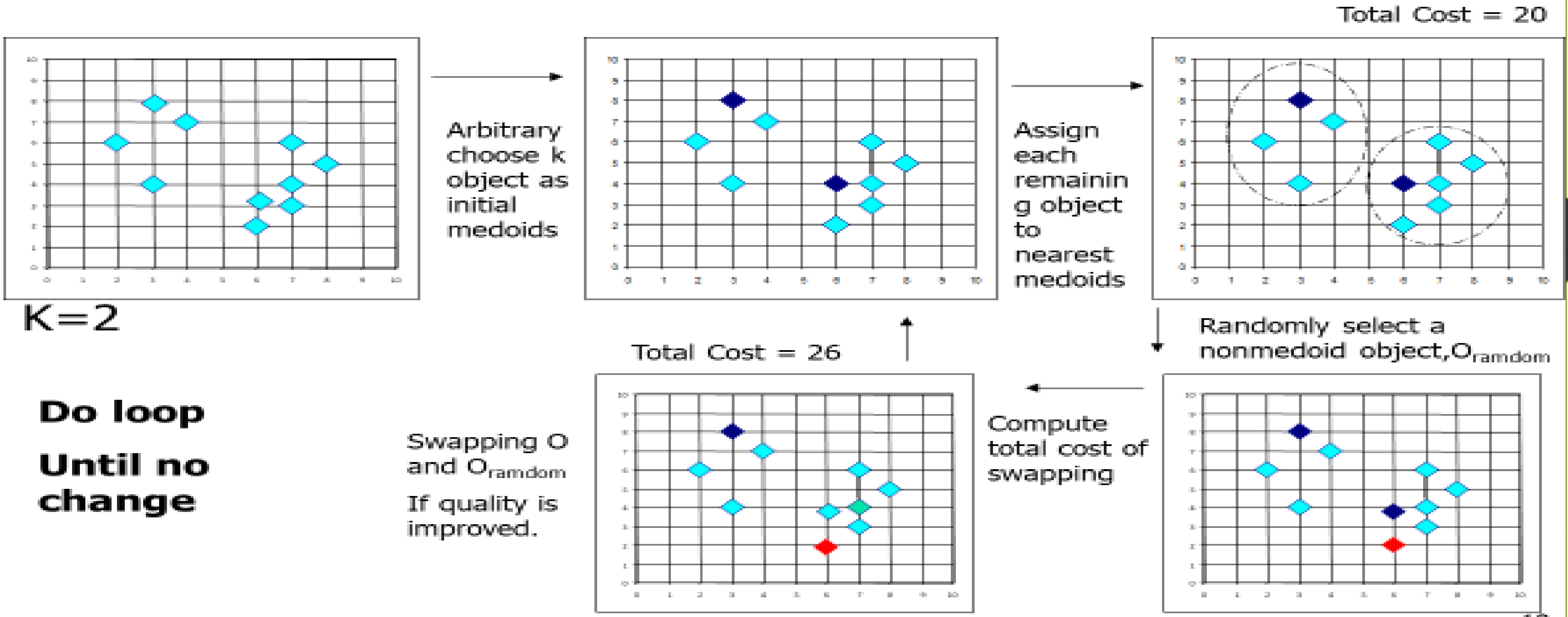
datapoint	Distance from v1	Distance from v2	Assigned center
a1(1,1)	1.05	3.78	v1
a2(2,1)	0.33	3.15	v1
a3(2,3)	1.67	1.8	v1
a4(3,2)	1.204	1.8	v1
a5(4,3)	2.605	0.75	v2
a6(5,5)	4.74	1.88	v2

- Cluster 1 of v1 ={a1,a2,a3,a4}
- Cluster 2 of v2={a5,a6}
- Recalculate the cluster centers:
- v1= (2,1.75)
- v2=(4.5,4)

Final clusters:

datapoint	Distance from v1	Distance from v2	Assigned center
a1(1,1)	1.25	4.61	v1
a2(2,1)	0.75	3.9	v1
a3(2,3)	1.25	2.69	v1
a4(3,2)	1.03	2.5	v1
a5(4,3)	2.36	1.12	v2
a6(5,5)	4.42	1.12	v2

PAM: A Typical K-Medoids Algorithm



K-Medoids Clustering

- K-Medoids Clustering: Find representative objects (medoids) in clusters
- PAM (Partitioning Around Medoids, Kaufmann & Rousseeuw 1987)
  - Starts from an initial set of medoids and iteratively replaces one of the medoids by one of the non-medoids if it improves the total distance of the resulting clustering
  - PAM works effectively for small data sets, but does not scale well for large data sets (due to the computational complexity)

K-Medoids:

1. Choose k number of random points from the data and assign these k points to k number of clusters. These are the initial medoids.
2. For all the remaining data points, calculate the distance from each medoid and assign it to the cluster with the nearest medoid.
3. Calculate the total cost (Sum of all the distances from all the data points to the medoids)
4. Select a random point as the new medoid and swap it with the previous medoid. Repeat 2 and 3 steps.
5. If the total cost of the new medoid is less than that of the previous medoid, make the new medoid permanent and repeat step 4.
6. If the total cost of the new medoid is greater than the cost of the previous medoid, undo the swap and repeat step 4.
7. The Repetitions have to continue until no change is encountered with new medoids to classify data points.

example

	x	y
0	5	4
1	7	7
2	1	3
3	8	6
4	4	9

- K=2
- Initial medoids: M1(1, 3) and M2(4, 9)
- Calculation of distances

Calculation of distance:

	x<	y	From M1(1, 3)	From M2(4, 9)
0	5	4	5	6
1	7	7	10	5
2	1	3	-	-
3	8	6	10	7
4	4	9	-	-

Cluster 1: 0

Cluster 2: 1, 3

- Calculation of total cost:

$$(5) + (5 + 7) = 17$$

- Random medoid: (5, 4)

M1(5, 4) and M2(4, 9):

	x	y	From M1(5, 4)	From M2(4, 9)
0	5	4	-	-
1	7	7	5	5
2	1	3	5	9
3	8	6	5	7
4	4	9	-	-

Cluster 1: 2, 3

Cluster 2:1

- Calculation of total cost:

$$(5 + 5) + 5 = 15$$

- Less than the previous cost
- New medoid: (5, 4).
- Random medoid: (7, 7)

M1(5, 4) and M2(7, 7)

	x	y	From M1(5, 4)	From M2(7, 7)
0	5	4	-	-
1	7	7	-	-
2	1	3	5	10
3	8	6	5	2
4	4	9	6	5

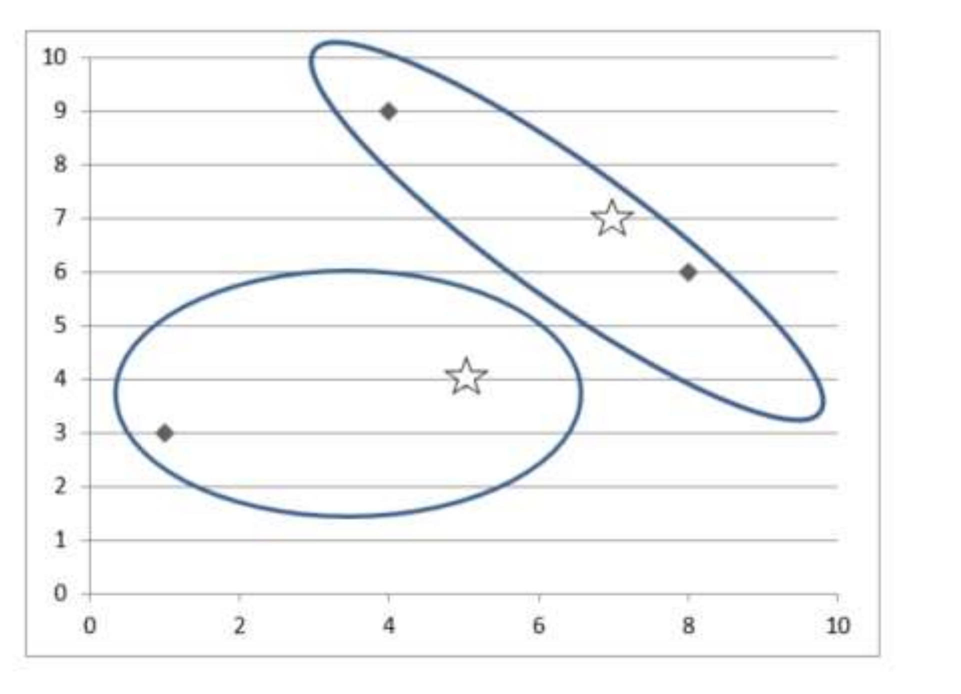
- Cluster 1: 2
- Cluster 2: 3, 4
- Calculation of total cost:
- (5) + (2 + 5) = 12
- Less than the previous cost
- New medoid: (7, 7).
- Random medoid: (8, 6)

M1(7, 7) and M2(8, 6)

	x	y	From M1(7, 7)	From M2(8, 6)
0	5	4	5	5
1	7	7	-	-
2	1	3	10	10
3	8	6	-	-
4	4	9	5	7

- Cluster 1: 4
- Cluster 2: 0, 2
- Calculation of total cost: (5) + (5 + 10) = 20
- **Greater than the previous cost**
- UNDO : Hence, the final medoids: M1(5, 4) and M2(7, 7)
- Cluster 1: 2
- Cluster 2: 3, 4
- Total cost: 12

Final results



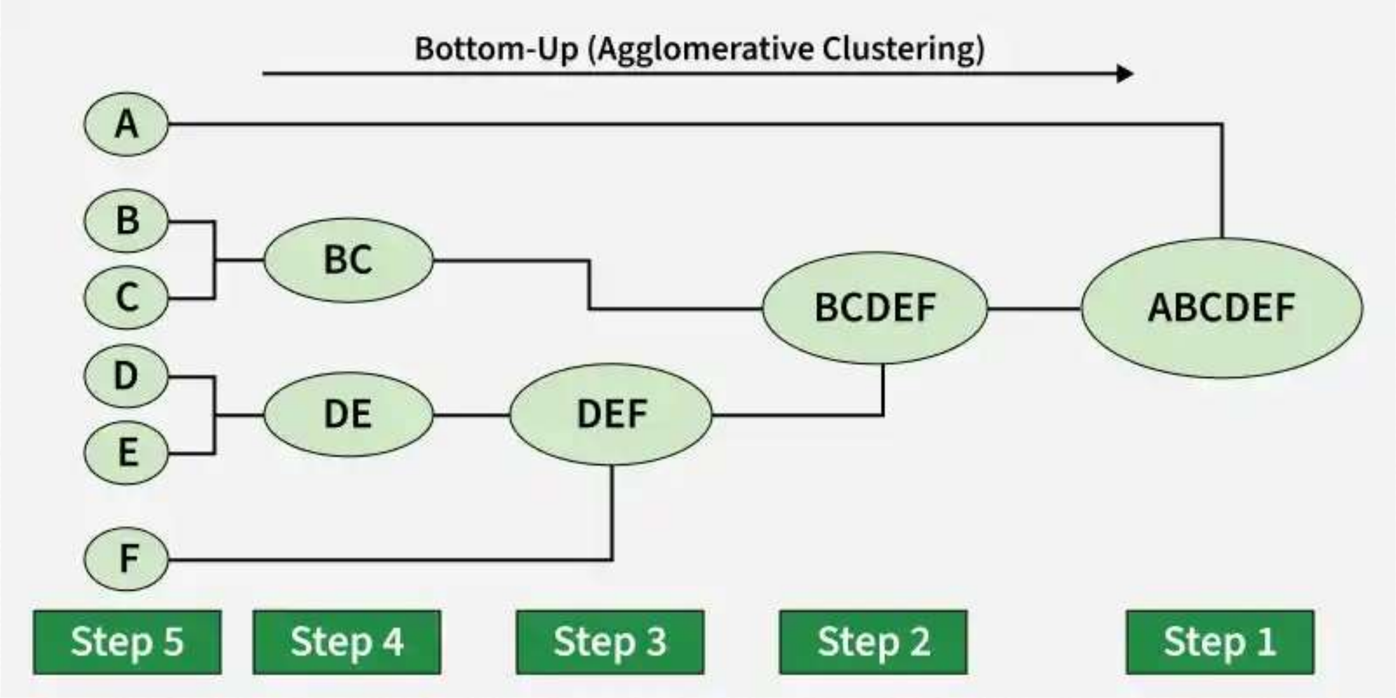
Hierarchical Clustering

- Hierarchical Clustering is an unsupervised learning technique that groups data into a hierarchy of clusters based on similarity.
- It builds a tree-like structure (dendrogram) that helps visualize relationships.
- There are two main types of hierarchical clustering.
- Agglomerative Clustering
- Divisive clustering

Agglomerative Clustering

- also known as the bottom-up approach or hierarchical agglomerative clustering (HAC).
- Bottom-up algorithms treat each data as a singleton cluster at the outset and then successively agglomerate pairs of clusters until all clusters have been merged into a single cluster that contains all data.

Agglomerative Clustering



Create a Dendogram ,Given a Distance Matrix

- Given the dataset {a, b, c, d, e} and

  the following distance matrix,

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Single Linkage Agglomerative Clustering Solved Example

- Dataset : $\{a, b, c, d, e\}$.
- Initial clustering (singleton sets)
- $\boldsymbol{C1}$: $\{\boldsymbol{a}\}, \{\boldsymbol{b}\}, \{\boldsymbol{c}\}, \{\boldsymbol{d}\}, \{\boldsymbol{e}\}$.
- From the table, the minimum distance is the distance between the clusters {c} and {e}.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

- Dataset : $\{a, b, c, d, e\}$.
- Initial clustering (singleton sets)
- $\boldsymbol{C1}$: $\{\boldsymbol{a}\}, \{\boldsymbol{b}\}, \{\boldsymbol{c}\}, \{\boldsymbol{d}\}, \{\boldsymbol{e}\}$.
- From the table, the minimum distance is the distance between the clusters {c} and {e}.
- Also, **d({c}, {e}) = 2.**
- We merge {c} and {e} to form the cluster {c, e}.
- The new set of clusters **C2: {a}, {b}, {d}, {c, e}.**

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a	b	c,e	d
a	0	9	?	6
b	9	0	?	5
c,e	?	?	0	?
d	6	5	?	0

- Let us compute the distance of **{c, e}** from other clusters.
- $d(\{c,e\},\{a\}) = min\{d(c,a), d(e,a)\} = min\{3, 11\} = 3$
- $d(\{c,e\},\{b\}) = min\{d(c,b), d(e,b)\} = min\{7, 10\} = 7$
- $d(\{c,e\},\{d\}) = min\{d(c,d), d(e,d)\} = min\{9, 8\} = 8$
- The following table gives the distances between the various clusters in C2.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a	b	c,e	d
a	0	9	3	6
b	9	0	7	5
c,e	3	7	0	8
d	6	5	8	0

- From the table, the minimum distance is the distance between the clusters **{a}** and **{c, e}.**
- Also, **d({a}, {c, e}) = 3.**
- We merge {a} and {c, e} to form the cluster {a, c, e}.
- The new set of clusters **C3: {a, c, e}, {b}, {d}.**

	a	b	c,e	d
a	0	9	3	6
b	9	0	7	5
c,e	3	7	0	8
d	6	5	8	0

	a,c,e	b	d
a,c,e	0	?	?
b	?	0	5
d	?	5	0

- Let us compute the distance of **{a, c, e}** from other clusters.
- $d(\{a,c,e\},\{b\}) = \min\{d(a,b), d(c,b), d(e,b)\}$
- $\boldsymbol{d(\{a,c,e\},\{b\}) \;=\; \{9,7,10\} \;=\; 7}$
- $d(\{a,c,e\},\{d\}) = \min\{d(a,d), d(c,d), d(e,d)\}$
- $\boldsymbol{d(\{a,c,e\},\{d\}) \;=\; \{6,9,8\} \;=\; 6}$

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a,c,e	b	d
a,c,e	0	7	6
b	7	0	5
d	6	5	0

- From the table, the minimum distance is between **{b}** and **{d}.**
- Also, d({b}, {d}) = 5.
- We merge {b} and {d} to form the cluster {b, d}.
- The new set of clusters **C4: {a, c, e}, {b, d}**

	a,c,e	b	d
a,c,e	0	7	6
b	7	0	5
d	6	5	0

	a,c,e	b,d
a,c,e	0	?
b,d	?	0

- $d(\{a,c,e\},\{b,d\}) =$

  $\min\{d(a,b), d(a,d), d(c,b), d(c,d), d(e,b), d(e,d)\}$

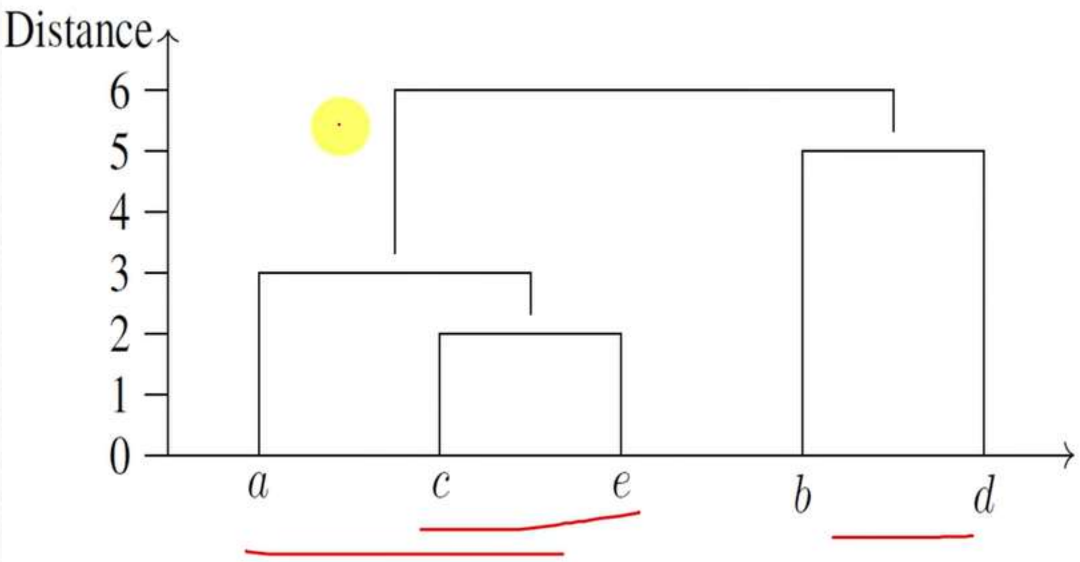
- $d(\{a,c,e\},\{b,d\}) = \min\{9,6,7,9,10,8\}$

- $\boldsymbol{d(\{a,c,e\},\{b,d\}) = 6}$

- Only two clusters are left. We merge them form a single cluster containing all data points.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a,c,e	b,d
a,c,e	0	6
b,d	6	0



	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Agglomerative Clustering

- Step – 1

	18	22	25	27	42	43
18	0	4	7	9	24	25
22	4	0	3	5	20	21
25	7	3	0	2	17	18
27	9	5	2	0	15	16
42	24	20	17	15	0	1
43	25	21	18	16	1	0

- Step – 1

	18	22	25	27	42	43
18	0	4	7	9	24	25
22	4	0	3	5	20	21
25	7	3	0	2	17	18
27	9	5	2	0	15	16
42	24	20	17	15	0	1
43	25	21	18	16	1	0

(42, 43)

- Step – 2

	18	22	25	27	42, 43
18	0	4	7	9	24
22	4	0	3	5	20
25	7	3	0	2	17
27	9	5	2	0	15
42, 43	24	20	17	15	0

- Step – 2

	18	22	25	27	42, 43
18	0	4	7	9	24
22	4	0	3	5	20
25	7	3	0	2	17
27	9	5	2	0	15
42, 43	24	20	17	15	0

(42, 43), (25, 27)

- Step – 3

	18	22	25, 27	42, 43
18	0	4	7	24
22	4	0	3	20
25, 27	7	3	0	15
42, 43	24	20	15	0

- Step – 3

	18	22	25, 27	42, 43
18	0	4	7	24
22	4	0	3	20
25, 27	7	3	0	15
42, 43	24	20	15	0

(42, 43), ((25, 27), 22)

- Step – 4

	18	22, 25, 27	42, 43
18	0	4	24
22, 25, 27	4	0	15
42, 43	24	15	0

- Step – 4

	18	22, 25, 27	42, 43
18	0	4	24
22, 25, 27	4	0	15
42, 43	24	15	0

(42, 43), (((25, 27), 22), 18)

- Step – 5

	18, 22, 25, 27	42, 43
18, 22, 25, 27	0	15
42, 43	15	0

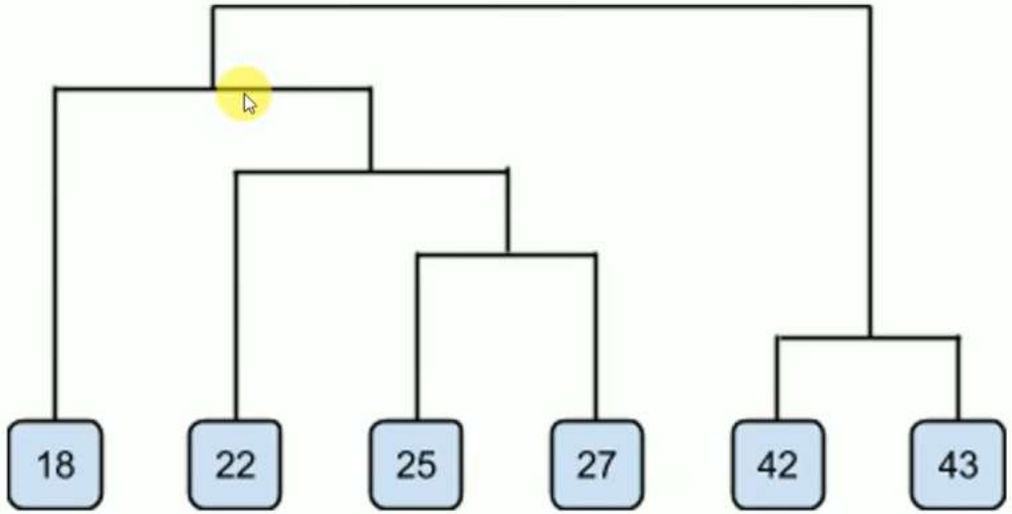
((42, 43), (((25, 27), 22), 18))

- Step – 6

	18, 22, 25, 27, 42, 43
18, 22, 25, 27, 42, 43	0

- **Dendrogram**

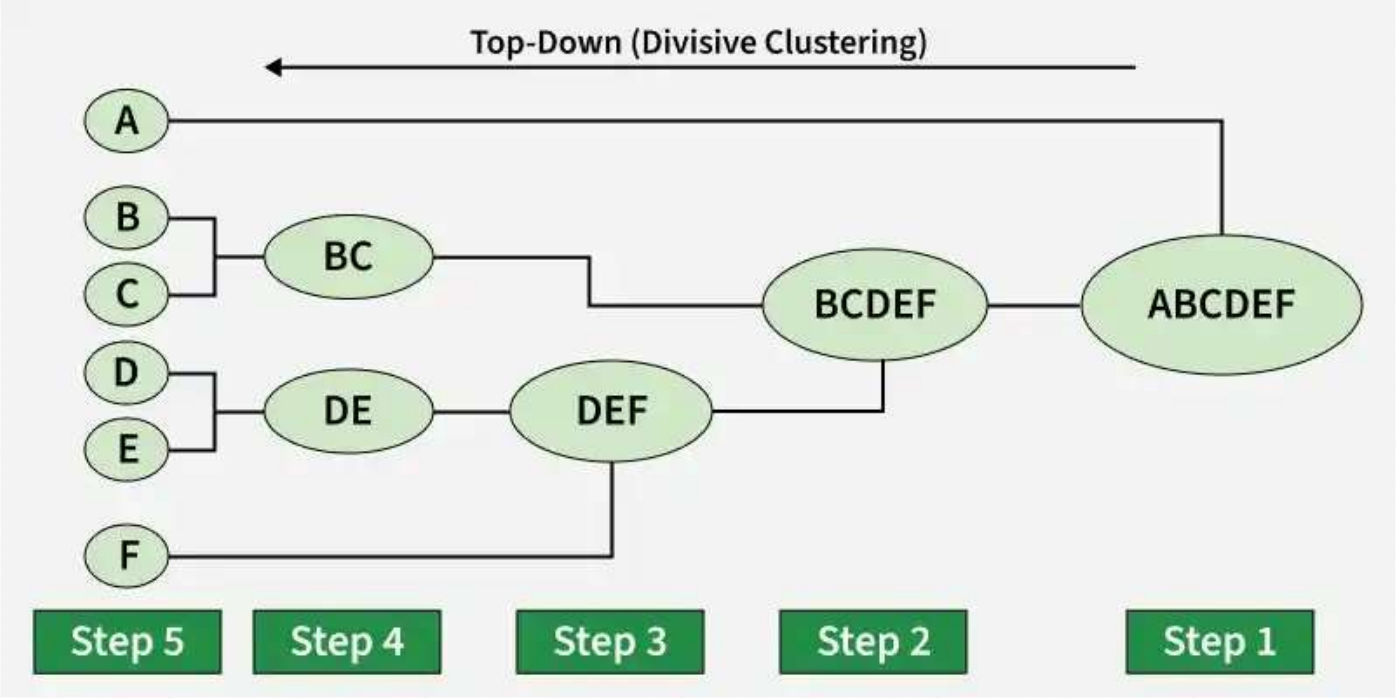
((42, 43), (((25, 27), 22), 18))



Divisive clustering

- Divisive clustering is also known as a top-down approach.
- Top-down clustering requires a method for splitting a cluster that contains the whole data and proceeds by splitting clusters recursively until individual data have been split into singleton clusters.

- **Start with all data points in one cluster:** Treat the entire dataset as a single large cluster.
- **Split the cluster**: Divide the cluster into two smaller clusters. The division is typically done by finding the two most dissimilar points in the cluster and using them to separate the data into two parts.
- **Repeat the process**: For each of the new clusters, repeat the splitting process: Choose the cluster with the most dissimilar points and split it again into two smaller clusters.
- **Stop when each data point is in its own cluster**: Continue this process until every data point is its own cluster or the stopping condition (such as a predefined number of clusters) is met.


Top-Down (Divisive Clustering)
A
B
C
D
E
F
BC
DE
DEF
BCDEF
ABCDEF
Step 5
Step 4
Step 3
Step 2
Step 1